

$$\S 1 \quad \frac{1}{f(\lambda)} e^{\lambda x} = \frac{1}{f(\lambda)} e^{\lambda x}$$

$$\int_1^{\infty} \frac{1}{x^p} dx \begin{cases} p > 1 \text{ 收敛} \\ p \leq 1 \text{ 发散} \end{cases} \quad \int_0^1 \frac{1}{x^p} dx \begin{cases} p < 1 \text{ 收敛} \\ p \geq 1 \text{ 发散} \end{cases}$$

$$\langle g | \hat{Q} f \rangle = \langle g | \hat{Q}^\dagger f \rangle$$

$$\beta = -i\hbar\alpha$$

§2

§A

A.1  $f(x) = x \sin(\frac{1}{x})$

~~不一致连续~~ 引例  $f(x) = x \sin(\frac{1}{x})$  在  $(0, 1]$  上连续

$g(x) = x^2$  在  $(0, \infty)$  上不一致连续  $x \rightarrow \infty$  时  $g(x) \rightarrow \infty$

A.2  $I_1 = \int_1^{\infty} \frac{\ln x}{1+x^2} dx$

收敛

$I_2 = \int_0^1 \frac{1}{\sqrt{1+x^2}} dx$

收敛

拆为  $\int_0^1 \frac{1}{\sqrt{1+x^2}} dx + \int_1^{\infty} \frac{1}{\sqrt{1+x^2}} dx$  收敛+收敛=收敛

§B  $B_1$   $p = q$   $y'' = p' = \frac{dp}{dx} = \frac{dp}{dy} \frac{dy}{dx} = p \frac{dp}{dy}$   $p \frac{dp}{dy} + p^2 = 1$

原方程  $p' - p^2 = 1$  初值  $y(0) = 1$   $y'(0) = 1$

$B_2$

$B_3$

$|A| = 0$

$(\lambda - 2)(\lambda - 1) = 0$

$\lambda = 2, \lambda = 1$

特征值  $\{1, 1, 1\} \rightarrow$  特征

$\{1, 1, 1\} \rightarrow$  特征值

$\{1, 2, 2\} \rightarrow$  特征值

$\{2, 2, 2\} \rightarrow$  特征值

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5C

$$\S 4 \quad I \quad \langle \psi | \hat{p} \psi \rangle = \langle \psi | -i\hbar \frac{d}{dx} \psi \rangle = \int_{-\infty}^{\infty} \psi^* (-i\hbar \frac{d}{dx} \psi) dx = (-i\hbar) \psi^* \psi \Big|_{-\infty}^{\infty} - (-i\hbar) \int_{-\infty}^{\infty} \frac{d\psi^*}{dx} \psi dx$$

$$= \int_{-\infty}^{\infty} i\hbar \frac{d\psi^*}{dx} \psi dx = \int_{-\infty}^{\infty} i\hbar \frac{d}{dx} \psi^* \psi dx = \int_{-\infty}^{\infty} (p\psi)^* \psi dx = \langle \hat{p} \psi | \psi \rangle = \langle \hat{p} \psi | \psi \rangle$$

$$II \quad \hat{C} = i[\hat{A}, \hat{B}] = i\hat{A}\hat{B} - i\hat{B}\hat{A}$$

$$\hat{C}^\dagger = -i\hat{A}\hat{B} + i\hat{B}\hat{A} = i[\hat{B}, \hat{A}]$$

$$\hat{C}^\dagger = -i[\hat{A}\hat{B} - \hat{B}\hat{A}] = -i[(\hat{B}^\dagger \hat{A}^\dagger)^\dagger - (\hat{A}^\dagger \hat{B}^\dagger)^\dagger] = -i(\hat{B}\hat{A} - \hat{A}\hat{B}) = i[\hat{A}, \hat{B}] = \hat{C}$$

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$$I \quad [\hat{x}, \hat{p}^2] = \hat{p}[\hat{x}, \hat{p}] + [\hat{x}, \hat{p}]\hat{p} = i\hbar \hat{p} + i\hbar \hat{p} = 2i\hbar \hat{p}$$

$$II \quad [\hat{x}, \hat{p}] = \hat{x}[\hat{x}, \hat{p}] + [\hat{x}, \hat{p}]\hat{x} = 0 + i\hbar \hat{x} = i\hbar \hat{x}$$

$$I \quad [\hat{x}, \hat{p}] = i\hbar$$

$$II \quad 2i\hbar(\hat{x}\hat{p} + \hat{p}\hat{x})$$

III

$$\Delta x \Delta p \geq \frac{1}{2} |\langle [\hat{x}, \hat{p}] \rangle| = \frac{1}{2} \hbar$$